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FUNDAMENTALS OF OPTIMIZATION

Metaheuristic methods

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CONTENT

- Introduction to Metaheuristics
- Tabu Search (TS)
- Simulated Annealing (SA)
- Genetic algorithms (GA)
- Ant Colony Optimization (ACO)

- Metaheuristics
 - Generic framework, problem-independent
 - Aim to find high-quality solutions to optimization problems in reasonable time
 - Do not specify (like heuristic methods) how close are the obtained solution from the optimal solutions
 - Single-solution based: Tabu Search, Simulated Annealing, Variable Neighborhood Search, etc.
 - Population-based: Genetic algorithms, Ant Colony Optimization, etc.

Tabu Search

- Based on neighborhood search
- Use memory structure to avoid revisiting solutions explored so far

```
function TabuSelect( $N$ ,  $f$ ,  $s$ ,  $T$ ){  
    eval =  $\infty$ ;  $s$  = NULL;  
    for  $x$  in  $N(s)$  do  
        if  $x \notin T$  and  $f(x) < \text{eval}$  then {  
            eval =  $f(x)$ ;  $s$  =  $x$ ;  
        }  
    return  $s$ ;  
}
```

```
function TabuSearch( $N$ ,  $f$ ){  
    //  $N$ : neighborhood  
    //  $f$ : quality function  
     $s$  = Generate an initial solution;  
     $s^*$  =  $s$ ;  
     $T$  = initialize a tabu list  
    while termination not reach do {  
         $s$  = TabuSelect( $N(s)$ ,  $f$ ,  $s$ ,  $T$ ); //neighbor selection  
        if  $s$  = NULL then break;  
        if  $f(s) > f(s^*)$  then  $s^*$  =  $s$ ;  
        AddLast( $s$ ,  $T$ ); // add  $s$  to the end of  $T$   
    }  
    return  $s^*$ ;  
}
```

Simulated Annealing

- Probabilistic technique based on neighborhood search
- Accept poor solutions with a probability which reduces over time

```
function S-Metropolis( $N, f, s, t$ ){  
    select  $n \in N$  with probability  $1/|N|$ ;  
    if  $f(n) > f(s)$  then return  $n$ ;  
    else with probability  $e^{\frac{f(s)-f(n)}{t}}$ ;  
    else return  $s$ ;  
}
```

```
function SimulatedAnnealing( $N, f$ ){  
    //  $N$ : neighborhood  
    //  $f$ : quality function  
     $s$  = Generate an initial solution;  
     $s^* = s$ ;  
     $t$  = initial templature;  
    while termination not reach do {  
         $s = S\text{-Metropolis}(N(s), f, s, t)$ ; //neighbor selection  
        if  $s = \text{NULL}$  then break;  
        if  $f(s) > f(s^*)$  then  $s^* = s$ ;  
         $t = \text{update}(t)$ ; // reduce temperature  
    }  
    return  $s^*$ ;  
}
```

Genetic algorithms

- Population-based method
- Generate initial population of solutions
- Perform cross over between 2 parent solutions from the population for generating new child solutions
- Perform mutation on child solutions with some probability
- Replace some poor solutions in the population by new good child solutions generated

Ant Colony Optimization (ACO)

- Nature-inspired metaheuristic
- Each iteration
 - m ants construct m solutions exploiting a pheromone model
 - Solution can be seen a sequence of components: component (i, j) means (for instance) the assignment of the decision variable x_i by the value j
 - Select component (i, j) with the probability

$$p(i, j) = \frac{\tau(i, j)^{\alpha} \eta(i, j)^{\beta}}{\sum_{(p, q) \in \text{Candidate}} \tau(p, q)^{\alpha} \eta(p, q)^{\beta}}$$

$\tau(i, j)$: pheromone on component (i, j)

$\eta(i, j)$: heuristic function for selecting component (i, j)

- Some obtained solutions will be used to update pheromone

- Example: $\tau(i, j) = \rho^* \tau(i, j) + \sum_{k=1}^m \Delta(i, j)^k$

in which $\Delta(i, j)^k = \begin{cases} \frac{1}{f(s^{best})}, & \text{if } (i, j) \text{ is selected in the solution constructed by ant } k \\ 0, & \text{otherwise} \end{cases}$

A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern of red dots. The dots are arranged in concentric, slightly irregular rings, creating a sense of depth and movement. In the center of this pattern, the word "HUST" is written in a bold, white, sans-serif font.

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THANK YOU !